

Jared F. Miller

CONVEX OPTIMIZATION · NONLINEAR SYSTEMS · CONTROL

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Education

Northeastern University

Boston, MA, USA

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING

Sept. 2018 - May 2023

- Communications, Control, and Signal Processing (CCSP)
- Advised by Prof. Mario Sznaiar
- Thesis Title: “Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures” [link]
- Thesis Committee: Octavia Camps, Didier Henrion (LAAS-CNRS), Bahram Shafai, Eduardo Sontag, Mario Sznaiar
- GPA: 4.0 (4.0 Scale)

M.S. IN ELECTRICAL AND COMPUTER ENGINEERING

Sept. 2015 - May 2018

- Communications, Control, and Signal Processing (CCSP)
- GPA: 3.852 (4.0 Scale)

B.S. IN ELECTRICAL ENGINEERING

Sept. 2013 - May 2018

- Minor in Mathematics
- GPA: 3.865 (4.0 Scale), Summa cum Laude

Research Publications

Journal Papers (published)

1. Jared Miller and Mario Sznaiar. Bounding the Distance to Unsafe Sets with Convex Optimization. *IEEE Transactions on Automatic Control*, 68(12):7575 – 7590, 2023. [link]
2. Jian Zheng, Tianyu Dai, Jared Miller, and Mario Sznaiar. Robust Data-Driven Safe Control using Density Functions. *IEEE Control Systems Letters*, 7:2611–2616, 2023. [link] (LCSS-CDC)
3. Jared Miller and Mario Sznaiar. Data-Driven Gain Scheduling Control of Linear Parameter-Varying Systems using Quadratic Matrix Inequalities. *IEEE Control Systems Letters*, 7:835–840, 2022. [link] (LCSS-ACC)
4. Jared Miller, Yang Zheng, Mario Sznaiar, and Antonis Papachristodoulou. Decomposed structured subsets for semidefinite and sum-of-squares optimization. *Automatica*, 137:110–125, 2022. [link]
5. J. Miller, D. Henrion, and M. Sznaiar. Peak Estimation Recovery and Safety Analysis. *IEEE Control Systems Letters*, 5(6):1982–1987, 2021. [link] (LCSS-ACC)
6. Jared Miller, Muhammad Ali Al-Radhawi, and Eduardo Daniel Sontag. Mediating Ribosomal Competition by Splitting Pools. *IEEE Control Systems Letters*, 5(5):1555–1560, 2021. [link] (LCSS-ACC)

Journal Papers (submitted)

1. Jared Miller, Tianyu Dai, and Mario Sznaiar. Data-Driven Superstabilizing Control under Quadratically-Bounded Errors-in-Variables Noise, 2024. [link]
2. Jared Miller and Mario Sznaiar. Analysis and Control of Input-Affine Dynamical Systems using Infinite-Dimensional Robust Counterparts, 2023. [link]
3. Jared Miller, Matteo Tacchi, Mario Sznaiar, and Ashkan Jasour. Peak Value-at-Risk Estimation of Stochastic Processes using Occupation Measures. 2024. [link]
4. Jared Miller, Matteo Tacchi, Didier Henrion, and Mario Sznaiar. Unsafe Probabilities and Risk Contours for Stochastic Processes using Convex Optimization, 2024. [link]
5. Jared Miller and Mario Sznaiar. Peak Estimation of Hybrid Systems with Convex Optimization, 2023. [link]
6. Jared Miller, Tianyu Dai, and Mario Sznaiar. Data-Driven Stabilizing and Robust Control of Discrete-Time Linear Systems with Error in Variables, 2023. [link]

Conference Proceedings (published)

1. Jared Miller, Tianyu Dai, and Mario Sznaiar. Superstabilizing Control of Discrete-Time ARX Models under Error in Variables. *IFAC-PapersOnLine*, 56(2):2444–2449, 2023. 22nd IFAC World Congress (Finalist for IFAC Young Author Award)
2. Jared Miller and Mario Sznaiar. Bounding the Distance of Closest Approach to Unsafe Sets with Occupation Measures. In *2022 61st IEEE Conference on Decision and Control (CDC)*, pages 5008–5013, 2022. [link]
3. Jared Miller, Tianyu Dai, and Mario Sznaiar. Data-Driven Superstabilizing Control of Error-in-Variables Discrete-Time Linear Systems. In *2022 61st IEEE Conference on Decision and Control (CDC)*, pages 4924–4929, 2022. [link] (Outstanding Student Paper Award)
4. Filip Bećanović, Jared Miller, Vincent Bonnet, Kosta Jovanović, and Samer Mohammed. Assessing the Quality of a Set of Basis Functions for Inverse Optimal Control via Projection onto Global Minimizers. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pages 7598–7605, 2022. [link]
5. Jared Miller and Mario Sznaiar. Facial Input Decompositions for Robust Peak Estimation under Polyhedral Uncertainty. *IFAC-PapersOnLine*, 55(25):55–60, 2022. [link] (IFAC Young Author Award)

Conference Proceedings (published)

6. Jared Miller, Didier Henrion, Mario Sznaier, and Milan Korda. Peak Estimation for Uncertain and Switched Systems. In *2021 60th IEEE Conference on Decision and Control (CDC)*, pages 3222–3228, 2021. [link] (Outstanding Student Paper Award)
7. J. Miller, R. Singh, and M. Sznaier. MIMO System Identification by Randomized Active-Set Methods. In *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 2246–2251, 2020. [link]
8. Jared Miller, Yang Zheng, Mario Sznaier, and Antonis Papachristodoulou. Decomposed Structured Subsets for Semidefinite Optimization. In *2020 21st IFAC World Congress*, 2020. [link]
9. Chieh Wu, Jared Miller, Yale Chang, Mario Sznaier, and Jennifer Dy. Solving Interpretable Kernel Dimensionality Reduction. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32, pages 7915–7925. Curran Associates, Inc., 2019. [link] (acceptance rate 21.9%)
10. J. Miller, Y. Zheng, B. Roig-Solvas, M. Sznaier, and A. Papachristodoulou. Chordal Decomposition in Rank Minimized Semidefinite Programs with Applications to Subspace Clustering. In *2019 IEEE 58th Conference on Decision and Control (CDC)*, pages 4916–4921, 2019. [link]
11. J. Miller and B. Shafai. A Model of Heave Dynamics for Bagged Air Cushioned Vehicles. In *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pages 976–981, 2019. [link]
12. B. Taskazan, J. Miller, U. Inyang-Udoh, O. Camps, and M. Sznaier. Domain Adaptation Based Fault Detection in Label Imbalanced Cyberphysical Systems. In *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pages 142–147, 2019. [link]

Conference Proceedings (accepted but not yet published)

1. Jared Miller and Roy Smith. Peak Estimation of Rational Systems using Convex Optimization, 2024. 2024 European Control Conference (ECC) [link]
2. Jared Miller, Jian Zheng, Mario Sznaier, and Chris Hixenbaugh. Data-Driven Superstabilization of Linear Systems under Quantization. In *2024 American Control Conference (ACC)*, 2024. [link]
3. Jared Miller, Matteo Tacchi, Mario Sznaier, and Ashkan Jasour. Peak Value-at-Risk Estimation for Stochastic Differential Equations using Occupation Measures. In *62nd IEEE Conference on Decision and Control*, 2023. [link]
4. Jared Miller, Tianyu Dai, Mario Sznaier, and Bahram Shafai. Data-Driven Control of Positive Linear Systems using Linear Programming. In *62nd IEEE Conference on Decision and Control*, 2023. [link]
5. Jared Miller, Milan Korda, Victor Magron, and Mario Sznaier. Peak Estimation of Time Delay Systems using Occupation Measures. In *62nd IEEE Conference on Decision and Control*, 2023. [link]

Conference Proceedings (submitted)

1. Mohamed Abdalmoaty, Jared Miller, Mingzhou Yin, and Roy S. Smith. Frequency-Domain Identification of Discrete-Time Systems using Sum-of-Rational Optimization, 2023. [link]

Preprints

1. Jared Miller, Chiara Meroni, Matteo Tacchi, and Mauricio Velasco. Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming, 2024. [link]
2. Jared Miller and Mario Sznaier. Quantifying the Safety of Trajectories using Peak-Minimizing Control, 2023. [link]
3. Tooba Imtiaz, Morgan Kohler, Jared Miller, Zifeng Wang, Mario Sznaier, Octavia Camps, and Jennifer Dy. SAIF: Sparse Adversarial and Interpretable Attack Framework, 2022. [link]

Seminars

1. “Data-Driven Safety Quantification using Robust Optimization,” February 16, 2024, Institute for Systems Theory and Automatic Control (IST), University of Stuttgart, Stuttgart, DE.
2. “Data-Driven Safety Quantification using Robust Optimization,” October 18, 2023, Cybernetic Systems and Controls Lab (CSCL), Arizona State University, Tempe, AZ. [link]
3. “Risk analysis for stochastic processes using polynomial optimization,” October 15-18, 2023, Convex Relaxations for Polynomial Optimization, INFORMS Annual Meeting, Phoenix, AZ [link].
4. “Data-Driven Safety Quantification using Robust Optimization,” October 11, 2023, Safe Autonomy and Intelligent Distributed Systems (SAIDS) group, University of Southern California, Los Angeles, CA. [link]
5. “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” July 27, 2023, Konstanz Real Algebraic Geometry Seminar, University of Konstanz. [link]
6. “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” May 29, 2023, Student Seminar Series on Optimization, Control & Learning, UC San Diego. [link]
7. “Quantifying Safety under Uncertainty using Occupation Measures,” May 26, 2023, Control Seminars @ UCI, UC Irvine
8. “Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization,” May 19, 2023, Multi-Robot Systems Lab Meeting, Stanford University. [link]
9. “Analysis and Control of Time-Delay Systems Using Polynomial Optimization,” May 14, 2023, MS14 Studying Dynamics using Polynomial Optimization Tools, SIAM Conference on Dynamical Systems. [link]
10. “Data-Driven Control under Input and Measurement Noise,” April 9, 2023, Oden Institute Seminar, UT Austin. [link]
11. “Safety Analysis for Nonlinear and Time-Delay Systems using Occupation Measures,” April 3, 2023, PhD Thesis Defense, Northeastern University. [link].
12. “Data-Driven Control under Input and Measurement Noise,” NYU MERIT Lab Seminar Series, New York City, Feb 21, 2023. [link]
13. “Bounding the Distance to Unsafe Sets with Convex Optimization”, DCSD Rising Stars, 2nd Modeling, Estimation and Control Conference, Jersey City, October 2-5 2022. [link]
14. “Bounding distances to unsafe sets”, June 16, 2022, IfA Coffee Talks, ETH Zurich. [link]

Seminars

16. “Bounding distances to unsafe sets”, June 14, 2022, LA3 Meeting, EPFL Lausanne. [link]
17. “Bounding distances to unsafe sets”, June 3, 2022, Journées SMAI MODE, University of Limoges (XLIM). [link]
18. Tutorials about Interior Point Methods, Polynomial Optimization, Frank-Wolfe algorithms and variations, and SDP approximations, May 16-20, Sparsity and Big Data in Control, Systems Identification, and Machine Learning, European Embedded Control Institute.
19. “Bounding distances to unsafe sets”, April 14, 2022, Conic Linear Optimization for Computer-Assisted Proofs, Mathematisches Forschungsinstitut Oberwolfach (MFO). [link]
20. “Bounding distances to unsafe sets”, June 28, 2021, Brainstorming days on measure and polynomial optimization (BrainPOP), LAAS-CNRS. [link]
21. “Data-Driven Peak and Reachability Set Estimation”, May 25, 2021, MS112 Methods of Learning Dynamical Systems for Control, SIAM Conference on Dynamical Systems. [link]
22. “Analysis and Control of Time-Delay Systems with Occupation Measures”, May 3, 2021, BrainPOP, LAAS-CNRS. [link]
23. “Exploiting Structure in Rank-Constrained and Approximated Semidefinite Programs”, December 19, 2019, TISEM Operations Research Seminar, Tilburg University. [link]

Seminars (upcoming)

1. “Maximizing the Slice-Volume of Semialgebraic Sets using Sum-of-Squares Programming,” March 11-14, 2024, Moments and Polynomials: Applications and Theory (MoPAT24), Konstanz, DE.
2. “Risk Analysis of Stochastic Processes using Auxiliary Functions,” April 11, 2024, Laboratory of Signals and Systems (L2S), Paris-Saclay, FR.

Poster Sessions (without Conference Proceedings)

1. “Risk Analysis of Stochastic Processes using Polynomial Optimization.” November 13, 2023, Future Trends in Polynomial Optimization, LAAS-CNRS, Toulouse, FR. [link]
2. “Frequency Domain Identification via Sum-of-Rational Optimization.” September 25, 2023, European Research Network System Identification (ERNSI), Stockholm, SE. [link]
3. “Safety Analysis and Control using Measures.” April 13, 2023, RISE 2023, Northeastern University. [link]
4. “Safety Analysis and Control using Measures.” February 27, 2023, PhD Research Expo, Northeastern University. [link]
5. “Diameter Constrained Minimum Spanning Graphs.” January 31, 2023, Current Themes of Discrete Optimization: Boot-camp for early-career researchers, ICERM. [link]
6. “Safety Analysis using Distance Estimation and Measures.” August 24, 2022. CLEVR-AI MURI Yearly Review Meeting, Northeastern University. [link]
7. “Exploiting SDP Structure Yields Tighter Approximations.” April 9, 2020. RISE, Northeastern University (remote). [link]
8. “Exploiting SDP Structure Yields Tighter Approximations.” February 24, 2020. IPAM Control, Learning and Optimization workshop, University of California, Los Angeles. [link]
9. “Chordal Decompositions in Rank Minimized SDPs.” May 30-31, 2019. Learning for Decision and Control (L4DC), Massachusetts Institute of Technology. [link]
10. “Chordal Decompositions in Rank Minimized SDPs.” May 10, 2019. New England Machine Learning Day, Northeastern University. [link]
11. “Scattered data interpolation through B-spline wavelets and the Elastic Net.” April 14, 2017. RISE, Northeastern University. [link]
12. “A parallelized Python-based Multi-Point Thomson Scattering analysis in NSTX-U.” October 29, 2014. 56th Annual APS Plasma Physics Conference, New Orleans. [link]

Experience

ETH Zürich

POSTDOCTORAL RESEARCHER, AUTOMATIC CONTROL LAB (IFA), FORSCHUNGSGRUPPE PROF. ROY SMITH

Zürich, CH

August 2023 - Present

Northeastern University

POSTDOCTORAL RESEARCHER, ROBUST SYSTEMS LAB, CONTROLS GROUP

Boston, MA, USA

May 2023 - July 2023

Northeastern University

RESEARCH ASSISTANT, ROBUST SYSTEMS LAB, CONTROLS GROUP

Boston, MA, USA

Jul. 2017 - April 2023

Laboratory for Analysis and Architecture of Systems (LAAS-CNRS)

CHATEAUBRIAND FELLOW, DECISION AND OPTIMIZATION, METHODS AND ALGORITHMS IN CONTROL (MAC) TEAM

Toulouse, FR

Jan. 2022 - Jul. 2022

Paradigm Hyperloop

CONTROL THEORIST, LEVITATION GROUP

Boston, MA, USA

Jul. 2017 - Dec. 2017

ASML Holding

CO-OP, METROLOGY GROUP

Veldhoven, NL

Mar. 2016 - Aug. 2016

Advanced Micro Devices (AMD)

CO-OP, SHADER COMPILER GROUP

Boxborough, MA, USA

Jan. 2015 - Jun. 2015

Princeton Plasma Physics Laboratory (PPPL)

INTERN/PART-TIME CONTRACTOR, DATA VISUALIZATION GROUP

Plainsboro Township, NJ, USA

Sep. 2012 - Feb. 2016

Cornell High Energy Synchrotron Source (CHESS)

SUMMER INTERN/RESEARCH ASSISTANT

Ithaca, NY, USA

Jul. 2012 - Aug. 2012

Teaching

ETH Zürich

TEACHING ASSISTANT

- Robust Control & Convex Optimization (227-0690-13L)

Zürich, CH

Spring 2024

TEACHING ASSISTANT

- System Identification (227-0689)

Fall 2023

Northeastern University

TEACHING ASSISTANT

- EECE 5644: Machine Learning and Pattern Recognition

Boston, MA, USA

Fall 2022

TEACHING ASSISTANT

- EECE 7345: Big Data, Sparsity, and Control

Fall 2021

European Embedded Control Institute

TEACHING ASSISTANT

- Sparsity and Big Data in Control, Systems Identification and Machine Learning

Toulouse, FR

May 16-20, 2022

Honors & Awards

Jul. 2023	Finalist for Young Author Award , 2023 IFAC World Congress	Yokohama, JP
Jun. 2023	Travel Award , 2023 American Control Conference (ACC)	San Diego, CA
Apr. 2023	ECE Excellence in Research Award , Northeastern University College of Engineering	Boston, MA
Jan. 2023	Travel Award , ICERM workshop: Current Themes of Discrete Optimization	Providence, RI
Dec. 2022	Outstanding Student Paper Award , 2022 61st IEEE Conference on Decision and Control	Cancún, MX
Dec. 2022	Travel Award , 2022 61st IEEE Conference on Decision and Control	Cancún, MX
Oct. 2022	ASME DSCD Rising Star Award , MECC 2022 (IFAC)	Jersey City, NJ, USA
Sep. 2022	IFAC Young Author Award , ROCOND 2022 (IFAC)	Kyoto, JP (remote)
Apr. 2022	Travel Award , MFO: Conic Linear Optimization for Computer-Assisted Proofs	Oberwolfach, DE
Jan. 2022	Travel Award (for Toulouse) , AFOSR FY22 International Student Exchange Program (ISEP)	Arvada, CO, USA
Dec. 2021	Outstanding Student Paper Award , 2021 60th Conference on Decision and Control	Austin, TX, USA
Apr. 2020	Chateaubriand Fellowship , Office of Science and Technology, Embassy of France in the USA	Toulouse, FR
Feb. 2020	Hosting , IPAM: Intersections between Control, Learning and Optimization (UCLA Workshop)	LA, CA, USA
Dec. 2019	Travel Award for Seminar , TISEM Seminar at Tilburg University	Tilburg, NL
Aug. 2019	Travel Award , 2019 IEEE Conference on Control Technology and Applications (CCTA)	Hong Kong
2013-2018	Honors Program , Northeastern University	Boston, MA, USA
2015-2018	Dean's List , Northeastern University	Boston, MA, USA

Skills

Programming	Matlab (incl. Simulink), Python, Mathematica, LaTeX, Julia, C/C++
MS Office	Word, Excel, PowerPoint, Publisher

Professional Organizations

Institute of Electrical and Electronics Engineers (IEEE)

MEMBER

Sept. 2013 - Present

IEEE Eta Kappa Nu (HKN)

MEMBER, TUTOR

Sept. 2014 - Present

Society for Industrial and Applied Mathematics

MEMBER

Oct. 2019 - Present

IEEE CSS Technical Committee on Robust and Complex Systems (TC-RoCS)

MEMBER

Sept. 2022 - Present

IEEE CSS Technical Committee on Hybrid Systems

MEMBER

Jun. 2023 - Present

IFAC Technical Committee 2.5 Robust Control

MEMBER

Sept. 2022 - Present

INFORMS

MEMBER

June 2023 - Present

Professional Service

Reviewer

Automatica, IEEE Transactions on Automatic Control (TAC), IEEE Control Systems Letters (L-CSS), Learning for Dynamics & Control Conference (L4DC), IFAC Symposium on Robust Control Design (ROCOND), IEEE Conference on Decision and Control (CDC), European Control Conference (ECC), Nonlinear Analysis: Hybrid Systems, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences; Kybernetika, Association for the Advancement of Artificial Intelligence (AAAI)